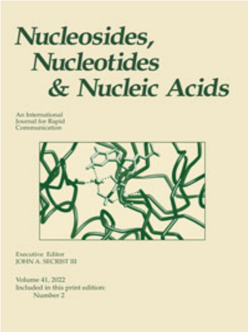

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Nucleosides, Nucleotides & Nucleic Acids

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/lncn20

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To cite this article: B. Vijila, E. H. Edinsha Gladis, N. Keerthi & J. Joseph (2022) Mesoporous aluminum impregnated rubber seed shell waste enriched with calcium as adsorbent material for the removal of microbial DNA in aqueous solution, Nucleosides, Nucleotides & Nucleic Acids, 41:2, 183-219, DOI: 10.1080/15257770.2022.2026382

To link to this article: https://doi.org/10.1080/15257770.2022.2026382

Published online: 12 Jan 2022.

NUCLEOSIDES, NUCLEOTIDES AND NUCLEIC ACIDS
2022, VOL. 41, NO. 2, 183–219
https://doi.org/10.1080/15257770.2022.2026382

Mesoporous aluminum impregnated rubber seed shell waste enriched with calcium as adsorbent material for the removal of microbial DNA in aqueous solution

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ABSTRACT
Water contamination by pathogens and diseases induced by these pathogens is a major water quality issue all over the world. Poor public health has been linked to tap water polluted with DNA harboring antibiotic resistance genes sequence. According to HSAB concept, surface modification of rubber seed shell waste with alumina (AIRSS) as novel agro-waste adsorbent creates more active surface constituents for DNA adsorption. The proximate, ultimate and EDAX analysis provides the percentage levels of ash concentration, volatile, moisture and fixed carbon content, elemental composition present in the adsorbent. The structural features of AIRSS were determined using FT-IR, SEM and XRD. In order to improve reaction conditions, the effect of pH, temperature, adsorbent amount, and reaction time is also examined. The highest percent of DNA removal (92.5%) was achieved at the optimum conditions: 2 g/L at pH 4, contact time 120 minutes as compared to the conventional methods. The DNA adsorbs onto the surface of AIRSS through physical (vander Waals force) and chemical interactions, as demonstrated by kinetics and spectroscopic analyses. Changes in enthalpy (H), free energy (G), and entropy (S) indicate that adsorption is a spontaneous and exothermic process, according to thermodynamic parameters. The results of the experiments showed that the prepared AIRSP adsorbent could be used to remove DNA from water. The efficacy of AIRSS for the removal of DNA has decreased after nine months of storage and use. Low pH and the presence of AIRSS improved DNA-AIRSS adsorption, according to our findings.

ARTICLE HISTORY
Received 26 May 2021
Accepted 3 January 2022

KEYWORDS
DNA; adsorbent; contaminated; adsorption; efficiency

1. Introduction

Extracellular DNA released by organisms can be found in both fresh and salt surface waters, as well as soils and sediments, and serves a variety of ecological functions. Antibiotic resistance genes (ARGs), which aid in the

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transmission and development of antibiotic-resistant bacteria, have emerged as emerging environmental contaminants in water,[1,2] necessitating a growing need to assess ARG levels and dangers in the environment. Antibiotic resistance genes found in extracellular DNA (eDNA) in treated wastewater effluents may lead to antimicrobial resistance (AMR) spreading in receiving waters. Research works have been performed using sophisticated therapeutics to eliminate cell-related antibiotic resistance genes (ARGs). It is generally known that eDNA can adsorb onto clay, suspended particles, and other soil components. This study looked into the probable and key mechanisms involved in DNA removal by adsorption onto modified adsorbent (AIRSS).

The utilization of agricultural waste products as novel adsorbent products to eliminate DNA from the contaminated source of water is due to its relative abundance, inexpensive, biodegradable, efficient and environmentally friendly disposal.[3,4] Instead of activated carbon, the present study was focused on inexpensive materials such as agricultural waste, in particular, rubber seed shell waste as an innovative adsorbent. Rubber seed shell waste (RSS) is biomass waste and currently converts into household items. This carbonaceous material i.e., adsorbent and is utilized for the extraction of DNA from water sources. Kanyakumari Dist, Tamil Nadu, India is famous for the cultivation of rubber (soil and environmental conditions support the growth and decide the elemental compositions). The biomass (obtained from massive amounts of RSS from oil factories as waste materials) is used to remove DNA molecules and replace the expensive industrial adsorbent.[5–7]

Due to its structural characteristics, the physico-chemical value (hardness) of Al^{3+} ($\eta = 45.77$) as hard acid and DNA (phosphate ion) as hard base. Therefore, the hard base DNA prefers to bind with aluminum ions on the basis of the HSAB concept. Since $Al_2(SO_4)_3$ is used in traditional water treatment, Al^{3+} is thought to be readily available in most developing countries. As a result, when looking for materials to defluoridate drinking water, Al^{3+} may be a good and long-lasting source of hard acid for binding DNA and then removing them from aqueous environments.

The DNA adsorbed onto the surface-modified AIRSS consists of (1) electrostatic interaction between phosphate ions on the outer layer of DNA and Al^{3+} ion and (2) ligand exchange between the adsorbent surface hydroxyl group and –OH groups of phosphate groups of DNA, respectively. Therefore, the adsorption depends on solution chemistry (pH, cation species, and concentrations). The surface-modified AIRSS is more hydrophobic has a lower proportion of hydroxyl groups and is enriched with Al^{3+} ion.

The aim of this investigations was to use aluminum compound impregnating to change the surfaces of RSS in order to establish hard surface sites for DNA adsorption from water or environmental sources in accordance with the HSAB principle. The suitability of RSS was achieved by the process of surface modification process and may reduce costs, provide long-term solutions for the removal of DNA. RSS is a natural polymeric material with a rough surface and porous structure that can provide more active surface sites with higher adsorption performance and excellent extracellular DNA removal efficiency. Batch adsorption experiments were used to investigate AIRSS's DNA adsorption efficiency.[8]

In the free-floating extracellular DNA (eDNA) fraction of microbes, antibiotic resistance genes (ARGs) and mobile genetic elements (MGEs) were identified. Due to natural transformation, these xenogenic components can produce bacterial cells that are resistant to one or more drugs. Because of the low concentration of eDNA extract in wastewater, getting good quality and extraction of eDNA was difficult. The present investigations focused on the adsorption approach for extracting eDNA from complicated wastewater mixtures without inducing uncontrollable cell lysis.

Based on the literature, rubber seed shell waste was chosen as an adsorbent for the elimination DNA removal. The adsorbent was characterized (FT-IR, TGA, EDX, XRD, SEM, BET) in order to better understand the nature of adsorption. The Langmuir and Freundlich adsorption isotherms were used to better understand the nature of the adsorption mechanism, while pseudo-first-order and second-order models were used to predict the nature of the adsorption kinetics. Using thermodynamic analysis, the viability, spontaneity, and existence of adsorption were investigated. Rubber-seed shell has been shown to be an appealing agro-waste material for chemical activation with aluminol to produce high-capacity activated carbon. The prepared adsorbents were compared to related adsorbents and their salient features were presented. As a result, this paper presents experimental findings on the preparation and activation of adsorbent made from rubber seed shell waste, as well as its application to DNA removal. UV-Visible spectroscopic information reveals that the aluminum specifically bound with guanine and adenine of N-7 position and also changes in wavelength at 260 nm. The latter spectral evidence clearly indicates that aluminum on the adsorbent surface selectively and specifically bound with DNA and hemicellulose, cellulose also interact with DNA through the intraparticle diffusion process.

2. Materials and methods

2.1. Adsorbent Preparation

The rubber seed was collected from the Kanyakumari District of Tamil Nadu, India. The unwanted soluble components, polysaccharides, hydrolyzable tannins, proteins, and colored components were removed from the collected rubber seed shell waste by washing it several times with hot water. Then it was thoroughly dried to remove the moisture content before being kept in a muffle furnace for 45 mins at 573 K. After being washed in distilled water, the ash was dried for 24 hours at 323 K. The ash was converted into the mesh size of 150 mm and stored in an airtight plastic container for future use and referred to as rubber seed shell waste (RSS). Analytical grade chemical reagents were used. Double distilled water was utilized in the experiments. The chemicals were supplied by Sigma–Aldrich Chemicals in India.

2.2. Aluminium impregnated rubber seed shell waste (AIRSS) preparation

50 g of RSS was combined with 0.3 M of 250 mL aluminum sulfate solution in a stirred reactor tank and thoroughly stirred. In the reactor, a 1 M sodium hydroxide solution was stirred at 200–300 rpm. Aluminum hydroxide is formed when NaOH reacts with $Al_2(SO_4)_3$, and it is then deposited on rubber seed shell waste ash. During this level, the pH of the reaction mixture had to be carefully regulated, which necessitated the addition of NaOH. When the pH of the reaction media reached the target range of 4–6, the NaOH was turned off. The final adsorbent material contains the mixture of aluminum hydroxide impregnated tea waste ash (AIRSS) and sodium sulfate. To obtain the desired adsorbent, the entire slurry was filtered and dried at 380 K. Before being dried in the oven for future use, the AIRSS was thoroughly washed in double distilled water to remove all sodium sulfate.

2.3. Characterization

The ash content, moisture, and volatile matter of the adsorbent were calculated using proximate measurement. The difference between the original total mass adsorbent and the sum of the mass of volatile, ash, and moisture was used to measure the amount of fixed carbon. Weighing 1.5 g of adsorbent sample and placing it in a weighted crucible in a preheated oven at 320 K for 3 hours. The sample was cooled to room temperature using desiccators before being weighed again. The moisture content (MC) of the activated tea waste adsorbent was measured by Equation (1). The sample was then heated in a muffle furnace at 780 °C for 8 mins. The crucible was then weighted after cooling in desiccators. Then the volatile content (VC) of the AIRSS was calculated by using Equation (2).[9] After the above determination, Equation (3) was used to measure the percentage of ash content (AC) in the AIRSS sample.[10] Ultimate analysis indicates the weight percent of carbon, nitrogen, hydrogen and sulfur content present in the AIRSS sample.

$$\text{Moisture content}\,(\%) = \frac{W1 - W2}{W1} \times 100 \tag{1}$$

$$\text{Volatile content}\,(\%) = \frac{W1 - W2}{W1} \times 100 \tag{2}$$

$$\text{Ash content}\,(\%) = \frac{Ws}{W1} \times 100 \tag{3}$$

where W_1 is weight of the sample, W_2 is the weight of the sample after drying, and W_s is the weight of the ash.

Using Equation (4), the fixed carbon content was measured by difference:

$$\text{Fixed carbon content}\,(\%) = 100 - (\%VC + \%MC + \%AC) \tag{4}$$

FTIR spectra were utilized to identify the nature of functionalities in the adsorbent. In spectra, the percentage of transmittance (% T) was plotted against the wave number (cm^{-1}). The spectra were collected in the wave number range (4000–400 cm^{-1}) and it was equipped with single attenuated total reflectance system. The adsorbent's surface morphology was studied with the help of the SEM technique. The SEM image of before and after adsorption determines the morphological changes. XRD provides information about the crystal phase, structure, preferred crystal orientation (texture), and other structural parameters, such as crystallinity, strain, average grain size, and crystal defects. The crystalline grain size of the powdered sample was calculated by using CuKα radiation at 25 °C. The line broadening in the X-ray peak is used to estimate the average crystalline grain size. Using Debye–Scherrer equation, the average crystalline sizes can be calculated by

$$d_{\mathrm{XRD}} = 0.9\lambda / \beta \cos\theta \tag{5}$$

where λ is the wavelength, θ is the diffraction angle (radians), β is the full width at half the maximum value and D is the particle size.

2.4. DNA material

Calf thymus DNA was purchased from Sigma and used as supplied. The DNA stock solution was prepared using buffers with different concentrations. The DNA concentration in solution was determined using UV-Visible spectrophotometer (Shimadzu), and the final result of concentration was an average of duplicate measurements and was calibrated by the standardization plot method.

2.5. DNA adsorption on the AIRSS

An adsorption isotherm was obtained by preparing a series of calf thymus DNA solutions with initial concentrations ranging from 10 to 200 μg/mL in deionized water. In each experiment, 10 μL of the DNA solution was added into a 0.5 mL centrifuge tube with 0.2 mg of AIRSS in 10 μL of deionized water. In addition, 20 μL of phosphate buffer was also added. The mixture was mixed thoroughly for 30 mins and then continuously shaken at 25 °C for 5 hrs. The final solution was placed under a magnetic field for 2 min to separate the AIRSS from the supernatant liquid. The amount of DNA adsorbed on AIRSS was calculated from the desorption of DNA concentrations after adsorption. To study the DNA adsorption capacities of AIRSS influenced by the varying binding solutions, the amounts of adsorbed DNA by AIRSS were determined under different adsorption conditions (pH, temperature, adsorbent dosage, etc). For all the adsorption and desorption measurements, duplicate samples were run, and the final result was the average of such two samples.

2.6. Adsorption process

Using batch equilibration, the influence of various parameters on the adsorption potential of aluminum impregnated rubber seed shell waste adsorbent (AIRSS), including contact time, adsorbent dose, DNA concentration, pH, stirring time and temperature, was investigated. DNA solutions with the desired concentration (10–50 mg/L) are made from a stock sodium DNA solution. In 100 mL of DNA solution, 0.25 g of adsorbent was added, and the solution was shaken for 150 minutes at 300 rpm. Then, it was treated with chelating agent like EDTA, to remove the Na^+ ions from the adsorbent. The solution was taken out from the shaker filtered through a vacuum filtration system to separate the adsorbent material. DNA conductance was measured by the ion-selective electrode. The following equation was used to quantify the volume of DNA adsorbed:

$$qe = \frac{(Ci - Ce)V}{W} \tag{6}$$

where q_e is the amount of DNA adsorbed (mg/g), C_i and C_e are the initial and final concentration of DNA (mg/L), V is the volume of the solution (L), and W is the weight of the adsorbent (mg).

2.7. Adsorption isotherm

Under some conditions, adsorption isotherms are the equilibrium relationship between adsorbate concentrations in the liquid phase and on the adsorbent surface. Langmuir isotherm is most widely used isotherm for either liquid or gas adsorption. It was created using the kinetics of gas molecule condensation and evaporation at a unit solid surface. The Langmuir isotherm assumes that sorption takes place on a structurally homogeneous adsorbent surface and models sorption surface monolayer coverage.

In Langmuir isotherm, the non-linear equation can be presented as

$$q_e = \frac{q_m k_L C_e}{1 + k_L C_e} \tag{7}$$

The linear form of the equation is

$$C_e / q_e = \left(1 / q_m k_L\right) + \left(C_e / q_m\right) \tag{8}$$

The Freundlich model explains that the sorption occurs on a heterogeneous surface and multi-layer coverage of sorption surfaces.

In the Freundlich isotherm model, the non- linear equation was formulated as

$$q_e = k_f C_e^{1/n} \tag{9}$$

The linear form of the equation is

$$\log q_e = \log k_f + \left(1 / n\right) \log C_e \tag{10}$$

where q_e is the adsorption capacity at equilibrium (mg/g), q_m is the maximum adsorption capacity (mg/g), k_L is the Langmuir isotherm constant (L/mg), C_e is the DNA concentration at equilibrium (mg/L) and k_f is the sorption capacity (mg per g) and n is the adsorption intensity.

2.8. Adsorption kinetics

The adsorption kinetics for AIRSS was introduced, with the pseudo first-order and second-order adsorption kinetics being the two primary adsorption kinetics represented in Table 4 and Figure 6. Equation (11) can be written for pseudo first-order kinetics.

$$\frac{dq_e}{dt} = k_1 \left(q_e - q_t\right) \tag{11}$$

In Equation (11) integration and rearranging it becomes

$$\ln\left(q_e - q_t\right) = \ln - q_e k_1 t \tag{12}$$

2.9. Determination of iodine number and pH_{PZC}

A volumetric method was used to calculate the iodine number of carbons present in the adsorbent (iodometry technique). The pH of the point of zero charge, pH_{PZC}, was determined (the pH above which the complete surface of the carbon particles is negatively charged). In this experiment, 0.02 L of a 0.01 M NaCl solution was put in a jacketed titration vessel that was thermostated at room temperature, and nitrogen was bubbled through the reaction mixture to avoid CO_2 dissolution and stabilize the pH. Then, the pH was adjusted between 2 and 10 in the reaction mixture containing adsorbent (0.1 g) with the drop by drop addition of either HCl or NaOH. After 48 hours, the final pH was measured and plotted against the initial pH. The pH at which the final pH crosses the line pH_{final} = $pH_{initial}$ is taken as the pH_{PZC} of the given carbon.

2.10. Determination of acidic and basic surface contents

The Boehm's titration method was used to assess the acidic and basic surface contents of carbons (Boehm, 1994). Under a nitrogen atmosphere, the adsorbent (0.1 g) was combined with 50 mL of 0.02 M NaOH or 0.02 M HCl. The excess of base and acid was titrated with 0.02 M HCl and 0.02 M NaOH, respectively, after 1 hour of stirring and filtration. The basicity and acidity of the surface were measured (assumed that all the acidic and basic functionalities were neutralized with the use of hydrochloric acid and sodium hydroxide).

2.11. Quantification of lignin, cellulose and hemicellulose

The National Renewable Energy Laboratory approach was utilized to evaluate the quantification of hemicellulose, cellulose and lignin present in the prepared adsorbent material. In this experimental approach, the rubber seed shell waste was subjected to oil extraction in 20 g was thoroughly mixed with 75 mL of hexane. To complete three extractions, the reaction mixture was vigorously shaken for 5 minutes before being set aside for 25 minutes. The hexane was isolated from the biomass and evaporated at 80 °C after the last extraction. The biomass was dried and removed, then suspended in 4 percent wt H_2SO_4 and stirred for 1 hour at RT. The suspension was then refluxed for 4 hours before being purified and analyzed to determine the presence of hemicelluloses, cellulose, and lignin. The biomass was weighed after being washed in deionized water and dried for 24 hours at 100 degrees Celsius. 0.15 g of biomass was suspended in 3 mL of 75% wt sulfuric acid and stirred for 24 hours at 4 °C. The resulting solution was then mixed with deionized water until it reached a total volume of 100 mL before being held on reflux for 3 hours. The cellulose content of the suspension was measured after it was cooled and filtered. Finally, deionized water was used to rinse the filtered solids until they were dried for 24 hours at 105 °C and weighed. Insoluble lignin was measured using the difference between this weight and the respective ashes content obtained after putting the same filtering crucible into a furnace at 550 °C for 5 hours. The quantification of lignin, hemicellulose and cellulose was compared with the UV spectroscopic approach (serial dilution of known concentrations of lignin, hemicellulose and cellulose with extractives from rubber seed shell waste at a specific wavelength of 246, 284 and 324 nm, respectively).

2.12. Adsorbent characterizations

The functionalities on the active surface sites of AIRSS were identified using the KBr disk approach on FTIR Affinity-1 model (Shimadzu, wave number from 4000 to 400 cm^{-1}). The surface morphological features of AIRSS at a higher magnification with the use of an SEM analyzer. Using a CHNS analyzer, the elemental proportions of sulfur, nitrogen, hydrogen, carbon was measured (Elemento analyzer, CDRI Lucknow). The Porosimetry System Analyzer was used to determine the surface area, pore size and pore volume of the prepared AIRSS. The pHpzc was calculated using 25 mg of AIRSS in a pH-regulated 25 mL NaCl solution. The solution was then agitated for two days on an orbital shaker, and the final pH was measured and noted. The point of zero charge (pHpzc) of AIRSS was determined using the mass titration method.

2.13. Leaching analysis for safe disposal of AIRSS

The toxicity of adsorbent is determined using a leaching test protocol developed by the State Environmental Protection Agency and used to determine if the discharge of spent adsorbent is safe or hazardous depending on the existence of the weak or strong contact between adsorbent and DNA ion. Two extraction liquids were used in the leaching test: vinegar with a pH of 5.0 (liquid 1) and vinegar with a pH of 3 (liquid 2). Based on the material alkalinity, the extraction liquids used for each of the waste (spent) adsorbent materials (both with and without stabilization) were determined. The pH values of the spent AIRSS samples were calculated using a 1: 3 ratio in double-distilled water (spent AIRSS: distilled water). Liquid 1 was used for stabilized spent AIRSS samples with a pH of less than 5 (pH 5), and liquid 2 was used for spent AIRSS samples with a pH greater than 5 (pH 8). With bottles containing the extraction solvent liquid weighed 20 times as much as the spent AIRSS), the spent AIRSS samples were shaken for 18 hours. Glass fiber filters (0.5 m) were used to distinguish the solid and liquid phase species. The DNA content of the liquid extracts was measured and compared to EPA DNA levels to see if it was at the recommended level. Each leaching test was repeated several times (mostly triplicate) to determine reproducibility.

3. Results and discussion

3.1. Elemental composition

The current study was focused on the use of rubber seed shell waste as a novel adsorbent and possible feedstock (based on abundance, cheaper and environmentally friendly). Rubber seed cultivation and agro waste materials collection in Kanyakumari, Tamil Nadu, India. The elemental composition of biomass (RSS) is responsible for the adsorptive strength of health-hazardous ions from water sources. Soil fertility and healthy atmosphere determine the elemental composition of the biomass (RSS). Generally, a raw biomass with low adsorptive efficiency was compared to carbonized or chemically activated biomass due to the increased weight percentage of carbon or inorganic metal ions bind with hazardous ion, respectively. In the present case, the RSS was carbonized at 700° C and then treated with aluminol to form the aluminum impregnated rubber seed shell waste. The following is a method for systematically identifying and removing toxic DNA ions from aqueous solution: Table 1 provides the elemental compositions in terms of weight percentage before and after chemical activation of RSS. Based on the EDX analysis, the major elements of the adsorbent were shown as carbon and oxygen whereas hydrogen, sulfur, and nitrogen was present as minor components.[5–7] Before chemical activation, the carbon content was 52%, and after chemical activation, it was 69.4%. According to the literature, biomass materials must have a carbon content of 40–80% and act as adsorbents. The prepared AIRSS met the minimum requirements and was developed as an adsorbent in the current study. The weight percentage of carbon content increased after carbonization and chemical activation, owing to the release of gases (volatile and non-volatile). Inorganic elements, as well as oxygen, sulfur, and nitrogen-containing functional groups, are all present.[8]

Table 1. Elemental compositions comparison.

Elements	Raw RSS	Carbonized RSS
Carbon	52.0	69.4
Hydrogen	5.3	2.1
Nitrogen	1.6	3.0
Oxygen	36.5	22.3
Sulfur	0.1	0.1
Calcium	3.1	3.1
Total	100	100

Table 2. Proximate content present in the AIRSS.

Proximate analysis
Moisture content 6 %
Ash content 18%
Volatile content 16%
Fixed carbon content 60%

According to the HSAB principle, aluminol (AlOH)-based surface functionalization is a more effective approach for increasing the removal potential of negatively charged species. Such a chemical modification on the surface of RSS defence against DNA may result in a novel adsorbent with increased potency. Alumina with a high pH zero charge point (pHPZC 8.2) that allows for the adsorption of negatively charged particles such as DNA. The EPA has given its approval to the device, which uses activated alumina as a filter media for water purification.

The surface area of carbons with pore size larger than 1 nm was estimated using the iodine amount, which indicates the adsorption efficacy in micropores. With a specific surface area of 16.2 m^2/g, the iodine amount of AIRSS was determined to be 904 mg^2/g. The pHPZC of AIRSS and carbonized AIRSS was found to be 8.20 and 8.75, respectively. The inclusion of inorganic compounds during carbonization indicates the increased pHPZC against medium and is suitable for the removal of DNA from aqueous solutions.[9]

The moisture content of AIRSS is low (6%), suggesting that it is of high quality. The ash content and volatile matter were 16% and 18%, respectively, meaning that inorganic content was present in a very limited percentage. The percentage amounts of ash, volatile matter, moisture, and fixed carbon content are shown in Table 1. The high ash content is due to the adsorbent preparation process's use of low-quality raw materials. The processing of activated carbon was hampered by the high ash content, resulting in low carbon content, which can reduce adsorption ability and performance. As a result, scientists working on wastewater distillation processes using adsorption treatment technologies are concerned about developing adsorbents with low ash content. Ultimate analysis of AIRSS adsorbent shows the organic elemental composition. Table 2 shows that

Table 3. Chemical composition of rubber seed shell waste was utilized in the elimination of DNA ion from aqueous solution.

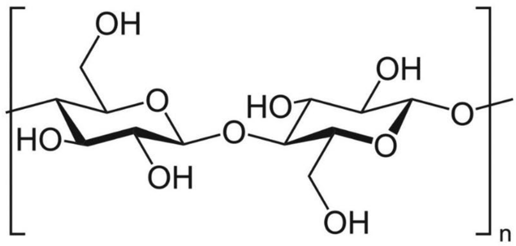
Chemical composition, %wt	AIRSS
Glucose, fructose and saccarose	3.50 ± 0.10
Hemicelluloses	58.61 ± 0.05
Cellulose	7.14 ± 0.07
Lignin	5.89 ± 0.06
Ash	22.42 ± 0.90
Unknown species	2.44 ± 0.71

high percentage of carbon content present in the adsorbent and sulfur present in very small percentage (<1). The EDS study showed that the adsorbent contains different proportions of C, O, Cl, Ca, and Al. The AIRSS contains about 45% impregnated aluminum and calcium ions. These metal ions are essential in the adsorption process and enhance the elimination of DNA ion from water sources.[10–13]

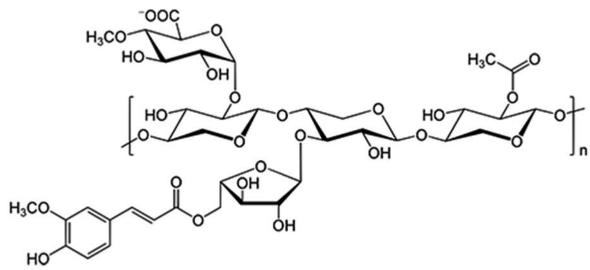
The natural biomass waste from rubber seed shell waste contained the basic constituents of cellulose, lignin, hemicellulose, sugar, and ashes. Very small quantities of other soluble organic and inorganic compounds were also present as nonstructural components.[14] The ash content of the prepared adsorbent (22.42%) indicated that it contained inorganic minerals such as K, Ca, Na, Mg, Si, and P, as well as oxygen.[15] The lignin in the checked adsorbent plays an important role in removing DNA ions (Table 3). Due to their effects on fiber swelling and water retention value, researchers have discovered that the contents of lignin and hemicellulose can affect the metal adsorption capacities of lignocellulosic fibers.

The chemical composition of biomass (e.g., cellulose, hemicellulose, and lignin content, Figure 1) is a key factor in determining adsorbent potency toward hazardous ions. Rubber seed shell waste, in particular, contains a higher percentage of lignin and thus has a higher adsorbent potency (Phenolic natural polymer underwent thermal decomposition at temperatures higher than those needed to degrade cellulose and hemicellulose molecules.[16] Hemicellulose, cellulose, and lignin, which were present on the surface of the adsorbents, are highly reactive at high temperatures.[17] These compounds degraded or decomposed due to the high temperature of carbonization (500 °C), resulting in structural collapse and fragmentation.[18] The adsorbent surface morphology supported the above observation.

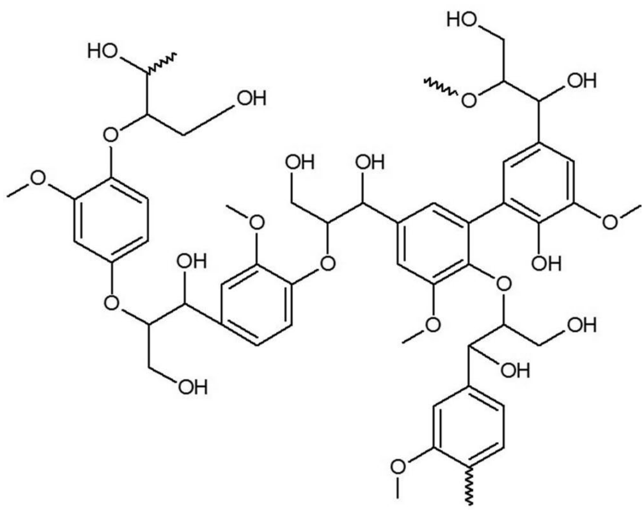
Porous carbon having small particle size, the rate of adsorption and diffusion are higher. Intra-particle diffusion decreases as particle size decreases due to the shorter mass transfer field, resulting in a higher degree of adsorption. AIRSS has a good removal performance since it was prepared in a powdered form. Manually measuring the particle



Structure of cellulose



Structure of Hemicellulose



Structure of lignin

Figure 1. Pictorial structures of cellulose, hemicellulose, and lignin.

size distribution was carried out. The particle size distribution measurement was carried out manually.[19,20] In this study, 75 mg of the sample was taken. The number of fine and coarse particles in this sample is determined after it was passed through various sieves. The data obtained for mesh size 52 is shown in Figure 2. The adsorbent's density has a significant impact on its adsorption capability. Literature data highlighted that the different densities of two adsorbents are

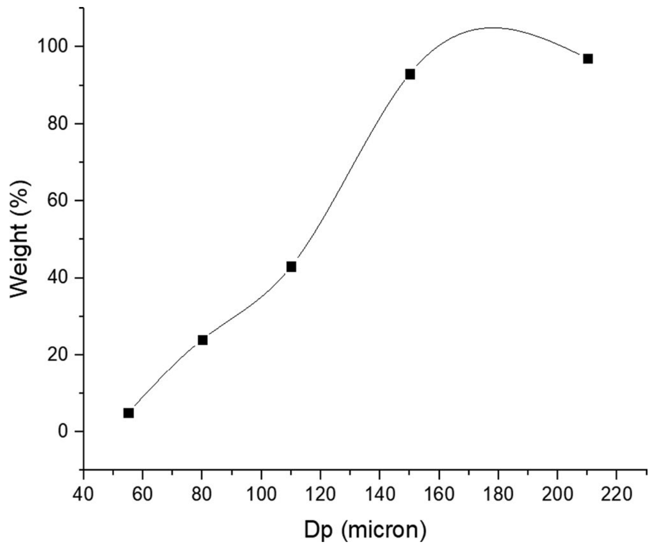


Figure 2. Particle size study by sieves for AIRSS.

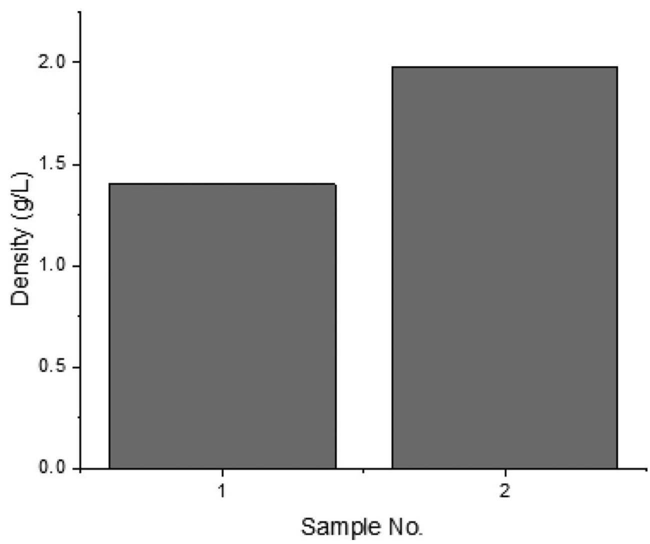


Figure 3. Density of AIRSS of different samples.

utilized at the same amount per liter; the highest density of adsorbents showed higher efficiency to extract the contaminants. The water displacement process can be used to quantify average bulk density. A certain quantity of adsorbent is used to measure the volume of water displaced in this process. Figure 3 shows the data of this experiment that indicate the average bulk density is 1.6525 g/mL. According to the findings, AIRSS had a total pore volume of 0.408 cm^3/g and a BET surface area of 515 m^2/g. This demonstrates that the proposed AIRSS

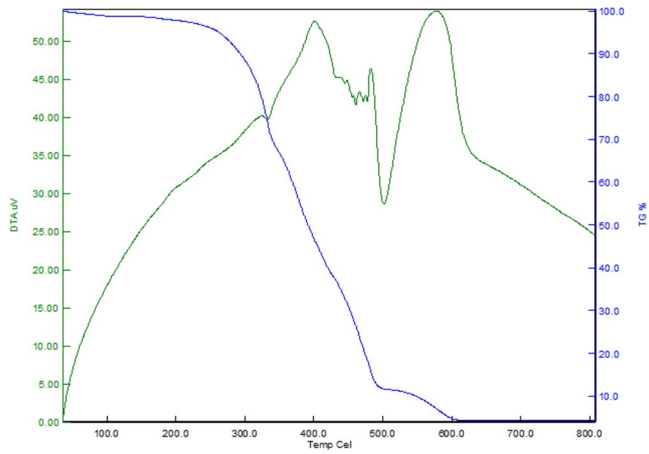


Figure 4. Thermogravimetric analysis of rubber seed shell waste.

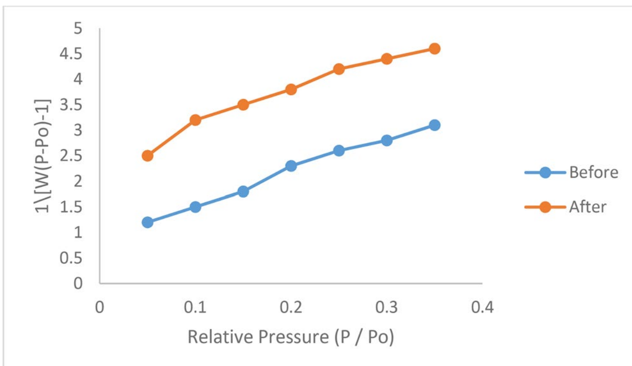


Figure 5. Brunauer, Emmett and Teller surface area plots where the magenta line is before DNA absorption and the navy blue line is after DNA adsorption.

is an effective adsorbent. Thermogravimetric analysis of the rubber seed shell waste (Figure 4) determines the stability and weight loss as a function of temperature, allowing the appropriate temperature for activation to be determined. The elimination of physisorbed moisture is responsible for the first peak of weight loss, which occurs between 50 and 150 degrees Celsius. Between 300 °C and 330 °C, active pyrolysis was observed, yielding a peak at 320 °C. It denotes hemicellulose decomposition and the release of volatiles from the natural material. The material becomes rich in carbon content at this stage, with rudimentary pores forming within the matrix. The final weight loss is 17% at 500 °C. In order to enhance the pore pathways and increase the surface area of activated carbon, a temperature of 500 °C is necessary.

Table 4. Determination of surface area, pore size and pore volume of the adsorbent before and after DNA adsorption.

Adsorbent	Surface area (m^2g^{-1})	Pore size (A°)	Pore volume (cm^3g^{-1})
Before DNA adsorption	422.5	14.60	0.420
After DNA adsorption	451.8	13.20	0.405

Table 5. Surface acidity and basicity of RSS and AIRSS.

Sample	Acidic (mmol g^{-1})	Basic (mmol g^{-1})
RSS	1.35	1.60
AIRSS	0.70	4.10

3.2. Bet analysis

BET measurements were performed using a fully automated analyzer to determine precise surface area,[21] pore size,[22] and pore volume[23] before and after DNA ion adsorption, and the results are summarized in Table 4. The surface area was measured assuming a monolayer of nitrogen adsorbed on the adsorbent. The isotherm plots were utilized to measure the surface area (BET approach/Nitrogen isotherm) and pore diameter of AIRSS and the surface features of the adsorbent are presented in Table 4. Adsorption may be occurred through multilayer adsorption, according to the above isotherms.

Using the BET equation in the p/po range of 0.05–0.35 (Figure 5), the surface area of AIRSS before and after DNA ion adsorption is 420.6 and 442.8 m^2g^{-1}, respectively. The decrease in pore size means that DNA are accumulating in the pores of AIRSS, as indicated by the increase in surface area. During the adsorption phenomenon's removal of DNA ions, the surface changes (chemical modification) in the nanoporous adsorbent's outer layer, resulting in an increase in the surface region.[24] Edge faces have a higher percentage of real surface area, so there are more reactive sites for DNA adsorption than intrinsic gibbsite.

The Dubinin–Astakhov (D–A) equation was used to determine the pore-size distributions and pore characteristics, and their plots are shown in Figure 6. Before and after DNA ion adsorption, the Dubinin–Astakhov pore radii are 13.40 A° and 11.60 A°, respectively. The decrease in pore radii is clearly evident due to DNA ion adsorption inside the pores of the adsorbent materials, which increases the surface area of the adsorbent. According to the findings of this analysis, pore radii decrease as surface area increases (Table 4).

3.3. Surface acidity/basicity

The acidic–basic character of carbons as calculated by Boehm titration is shown in Table 5. The acidity decreases and the basicity increases as the

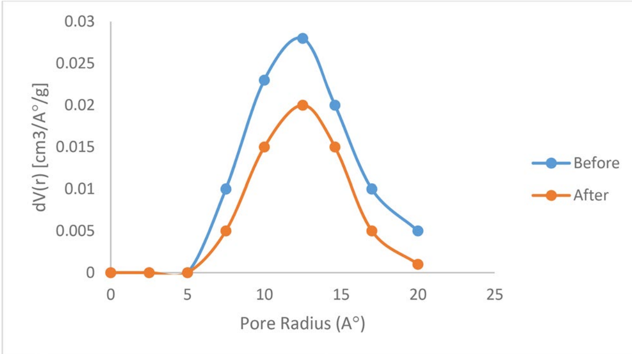


Figure 6. D–A plots for the pore radius of before DNA absorption and after DNA adsorption.

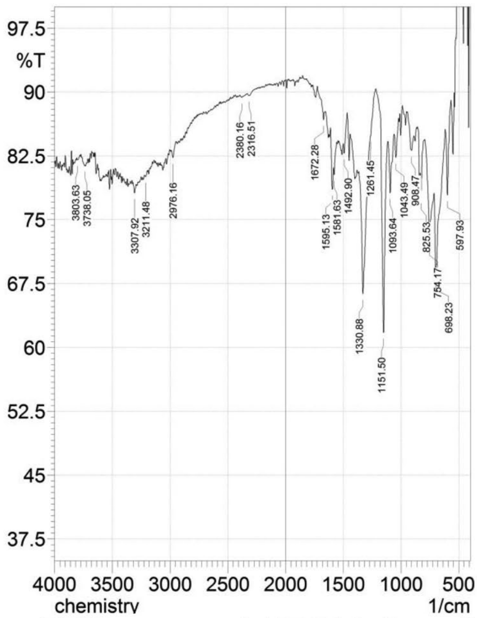
carbonization temperature rises. Carbons' high basicity, which exceeds 4.95 mmol g^{-1} for AIRSS at 700 °C, emphasizes their value. This finding can be explained by carbons' inorganic material, which allows CO_2 to be trapped during carbonization and the formation of carbonate compounds.[25]

3.4. FT-IR

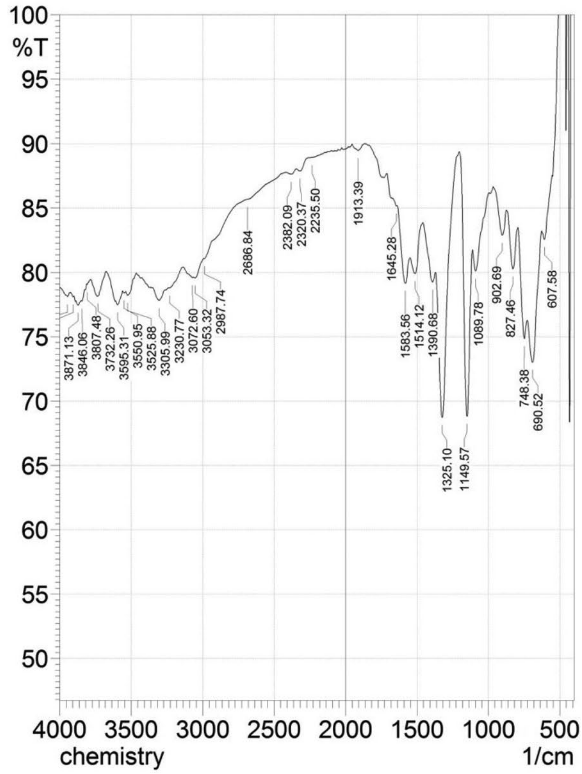
The functional group present in the tea waste adsorbent was identified by using FTIR spectroscopy. In spectra, the percentage of transmittance (% T) was plotted against the wave number (cm^{-1}). In the case of adsorbent alone (Figure 7a), the observed stretching vibrational frequencies at 3394, 1718, 1612, and 1047 cm^{-1}, which correspond to the stretching vibrations of the hydroxyl and C-O groups in carboxyl, C-O in $-OCH_3$, and –OH in alcohol, respectively.[25,26] A broadening of the band at 3390 cm^{-1} after DNA adsorption, which can be attributed to electrostatic adsorption between the adsorbent and the DNA.[27,28] After DNA adsorption, the vibrational band changes (Figure 7b) from 1625 to 1610 cm^{-1} due to the interaction of DNA ions with the hydroxyl groups on the adsorbent, resulting in a decrease in peak intensity. The -OH bending deformation in AIRSS is also visible in bands at 550–1,200 cm^{-1}. The treated material's spectra revealed new adsorption bands between 800 and 1,700 cm^{-1}, which corresponded to the Al-based OH mode.[29–31]

3.5. Powder XRD

The X-Ray Diffraction investigations gave the information about the particle size and nature (crystalline or amorphous) of the adsorbents before and after adsorption.[32] The diagram was noted at 2θ range from 10° to



a: FTIR spectra of AIRSS before adsorption



b: FTIR spectra of AIRSS after adsorption

Figure 7. (a) FTIR spectra of AIRSS before adsorption (b) FTIR spectra of AIRSS after adsorption.

80° and diffraction patterns were presented. In the case of adsorbent with before adsorption, eleven different peaks with 2θ values, 78.245°, 38.189°, 35.888°, 34.512°, 32.113°, 29.943°, 26.678°, 24.248°, 19.457°, 14.816° and 14.212° which relates to the inter-planar spacing (relative intensity) 1.22 (11.1%), 2.35 (16.5%), 2.50(15.6%), 2.6(14%), 2.79(24.2%), 2.98(14%), 3.34(48.5%), 3.67(91.7%), 4.56(100%), 5.97(70%) and 6.23 (50.8%). In the case of adsorbent with after adsorption, seven different peaks with 2θ values were 49.837°, 33.545°, 31.218°, 29.185°, 23.515°, 17.216° and 13.997° which relates to the inter-planar spacing (relative intensity) 1.83 (30.9%), 2.67 (24.2%), 2.87(23.8%),3.06(29.9%), 3.78 (100%),5.15(52.1%) and 6.32(32.5%) . The average crystalline size of the adsorbent with before and after adsorption was 61 and 52 nm was calculated. The XRD diagram of the adsorbents before and after adsorption was shown in Figures 8a and 8b.

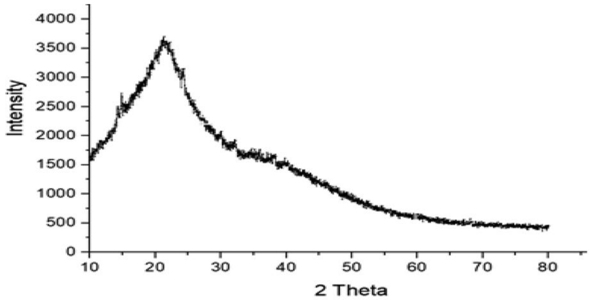
The crystalline structure of aluminum metal was studied using XRD analysis of AIRSS before and after adsorption (Figures 8a and 8b). For aluminum impregnated rubber seed shell wastesamples, there were no distinct peaks, implying that the aluminum on the rubber seed shell wastesurface was amorphous. Amorphous materials have a greater basic surface area and more active areas, making them effective adsorbents.[33] As a result, impregnated rubber seed shell waste ash was thought to be a strong biosorbent for DNA elimination.

3.6. SEM

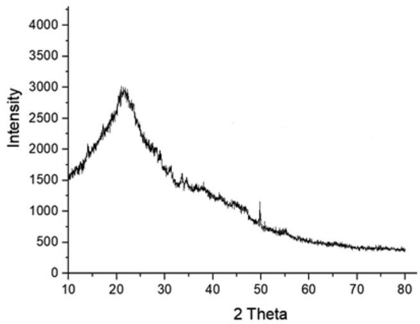
The AIRSS was subjected to SEM analysis, which revealed the adsorbent's surface texture and morphology. The higher surface area and efficiency of adsorption on the AIRSS are due to its porous nature and mostly microporous character. The maximum adsorption sites are provided by the adsorbent's surface roughness and porosity (ATWA), which promote DNA adsorption. The composition of the adsorbent particle changed after DNA adsorption. The surface texture of DNA on AIRSS was determined using SEM analysis. Before DNA adsorption, the surface was homogeneous and porous, as seen in the SEM image (Figure 9a), but after simultaneous adsorption of DNA, the surface was rough and less porous than before adsorption. Figure 9b shows that these pores were not present after adsorption, and the surface was rougher and bulkier than before adsorption.

3.7. Kinetic isotherms

The kinetics of any sorption mechanism is determined by a variety of factors, including adsorbent–adsorbate interactions, adsorbent's structural



a: XRD spectrum of AIRSS before adsorption



b: XRD spectrum of AIRSS after adsorption

Figure 8. (a) XRD spectrum of AIRSS before adsorption (b) XRD spectrum of AIRSS after adsorption.

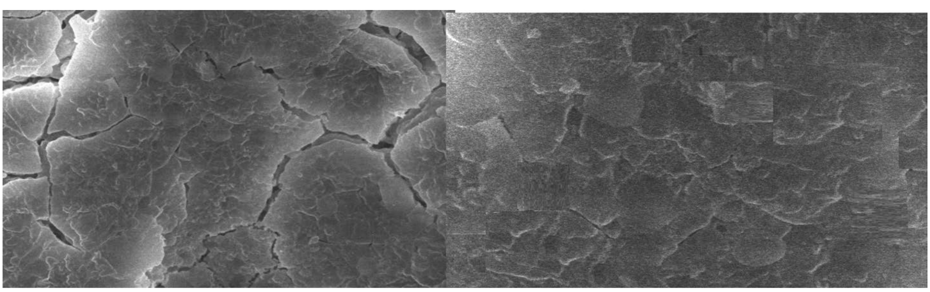
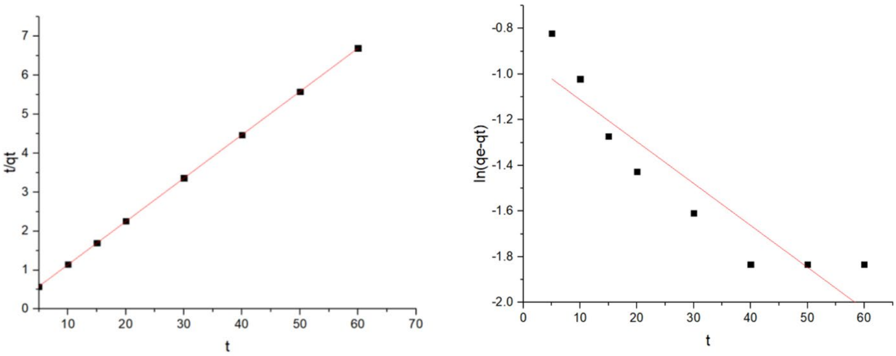


Figure 9. SEM image of AIRSS (a) before adsorption (b) after adsorption.

properties, concentration and nature of the adsorbate. The kinetic models are shown in Equations (5) and (6) and the output constant values of the kinetics are summarized in Table 5. Based on experimental (qe,exp) and theoretical evidence, the pseudo-first-order model agrees that rate change is related to the difference in equilibrium concentration with respect to time (qe,cal). Table 6 shows that the R^2 for AIRSS was 0.83195 by pseudo first order. The results showed that the AIRSS was not the right match for first-order because the experimental and theoretical evidence did not

Table 6. Kinetics and isotherms parameters for adsorption of DNA.

Pseudo first order	Pseudo second order	Langmuir model	Freundlich model
$K_1$0.00031	$K_2$0.455301	q_{max} 73.90983 K_L0.019204	n 1.21226 K_f1.696407
R^2 0.83195	R^2 0.99999	R_L 0.510148 R^2 0.99051	R^2 0.95259



a: Pseudo first order kinetics **b: Pseudo second order kinetics**

Figure 10. (a) Pseudo first order kinetics (b) Pseudo second order kinetics.

align well (Figure 10a). As a result, AIRSS adsorption does not obey, and the observed findings were fitted with a pseudo second-order model[24] (Figure 10b). The slope and intercept for the graph of t vs t/qt are qe and k^2. The R^2 was 0.99999, which was closer to 1 than the pseudo first-order model, indicating that the pseudo second-order model suited the data better than the pseudo first-order model.

In the Langmuir model, the graph was plotted against $1/C_e$ versus $1/q_m$, established a straight line and slope indicating the DNA adsorption on AIRSS shown in Figure 11a. The Longmuir constants k_L, R^2 and q_m calculated using the plot's intercept and slope, respectively. In the Freundlich model, the values of intercept and slope were obtained from the graph (log C_e versus log q_e) shown in Figure 11b. The correlation coefficient values of 0.99051 and 0.95259 are obtained from Langmuir and Freundlich models. The Langmuir adsorption isotherm model for DNA adsorption on AIRSS was found to be correct based on the experimental findings.

The intra-particle diffusion expression was established by Weber–Morris, where k_{id} is the intra-particle diffusion constant (mg $g^{-1}min^{-1(1/2)}$) and C the intercept. The plot of q_t vs $t_{1/2}$ gave a linear relationship and the values of both k_{id} and C were determined from the slope and intercept, respectively from Figure 12. Intercept C provides the thickness of the boundary layer, i.e., the resistance to the external mass transfer.

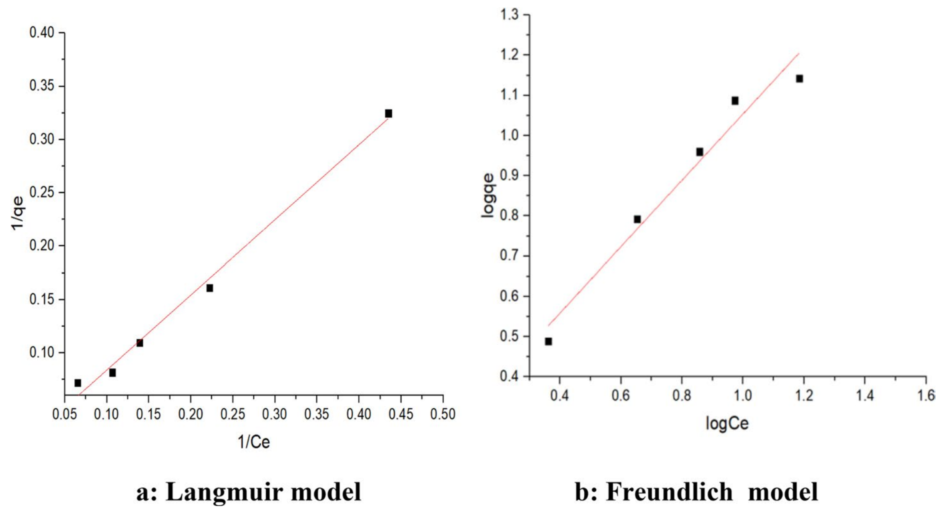


Figure 11. (a) Langmuir model (b) Freundlich model.

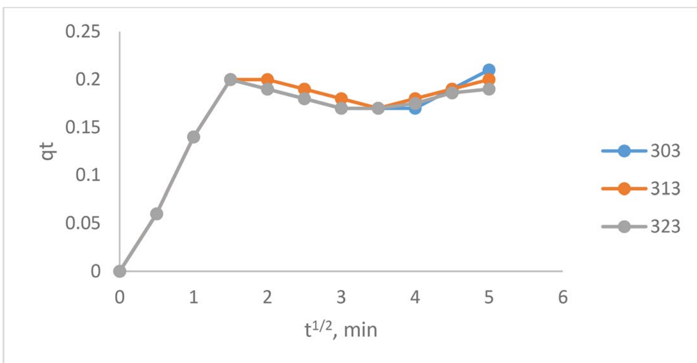


Figure 12. Weber–Morris (intra-particle diffusion) plot of DNA adsorption by AIRSS.

The Polanyi theorem-based Dubinin-Radushkevich (D-R) isotherm model can provide information about the energy needed for the adsorption process and mechanism. Equation (13) may be used to calculate the mean free energy of adsorption (E) for the DNA-AIRSS method, the magnitude of which can provide useful information about the adsorption mechanism, whether it is a chemical or physical interaction.

$$E = 1/\left(2K_{DR}\right)^{1/2} \tag{13}$$

The typical bonding energy range for an ion-exchange system has been stated to be 8–16 kJ/mol. It's also worth noting that physisorption occurs up to −22 kJ/mol due to electrostatic interaction between charged particles, while chemisorption occurs at negative values greater than −40 kJ/mol.

Physical adsorption plays a significant role in DNA uptake by AIRSS, according to the E value (35 kJ/mol) obtained in this analysis. The bonding energy of DNA and AIRSS is represented by b_T, which indicates the form of interaction between DNA and AIRSS. The Temkin isotherm constants are 0.860, 2.640, 2.271, and 0.956, which were calculated from the slope and intercept of a plot of qe versus Ce using the linear equation and provide B_T, b_T (kJ/mol), AT, and R^2. The order of magnitude of the energy value (b_T =2.271 kJ/mol) indicates that physisorption is involved in the interaction between DNA and AIRSS, which is consistent with the D-R isotherm predictions.[34,35]

In order to find out the optimum condition for the elimination of DNA from water was influenced by the different physico-chemical parameters as follows:

3.8. pH

At the water-adsorbent interface, the pH of an aqueous solution plays an important role in regulating the adsorption activity of various inorganic adsorbents. The initial solution pH is one of the most critical parameters in most adsorption processes, as it can greatly affect the degree of adsorption as well as determine the adsorption mechanism. This may be due to the fact that pH has an effect not only on the adsorbent's surface charge, but also on the adsorbate's speciation and degree of ionization. Batch adsorption experiments with solutions containing DNA concentrations of 9.0 mg/L were carried out to investigate the impact of pH on adsorption. Figure 13 shows that DNA adsorption potential increases at low pH and decreases at higher pH due to the formation of hydrofluoric acid in an acidic medium, causing the optimal adsorption to be lower than the neutral pH. At pH 4, with an initial DNA ion concentration of 2 mg/L, contact duration of 2 hours, and an adsorbent dose of 1 g/L, optimal DNA adsorption occurs. In an alkaline medium, OH^- and PO_4^- ions are repelling each other thus the DNA removal efficiency decreases. The result shows the maximum removal efficiency was 92.5% at pH 4.[34] DNA ions are negatively charged and are attracted to the positively charged adsorbent, resulting in an increase in adsorption power.[18,36]

Adsorbent (AIRSS) surface generally showed a negative charge at pH values higher than 3, and DNA molecules are also negatively charged due to phosphate groups when pH is above the isoelectric point (5) according to Cai et al.[37] and Pietramellara et al.[38] Therefore, at lower pH (4.0), both AIRSS surface and phosphate groups of DNA molecules are less negatively charged, so the repulsion between biochar and DNA decreased and, consequently, the adsorption increased. However, the effect of pH on

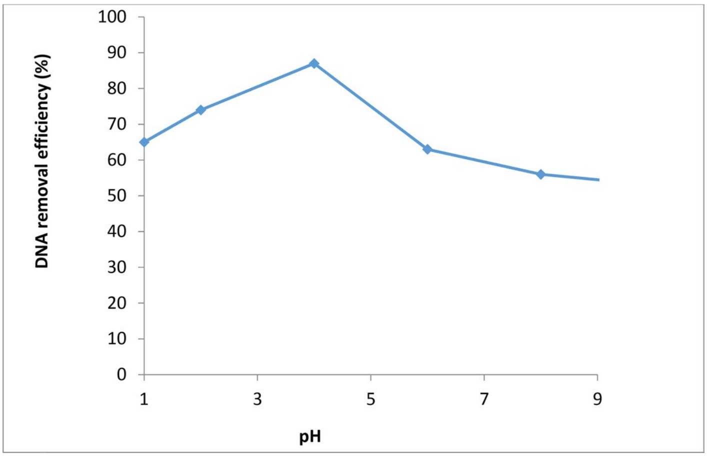


Figure 13. Effect of pH on DNA adsorbed on AIRSS.

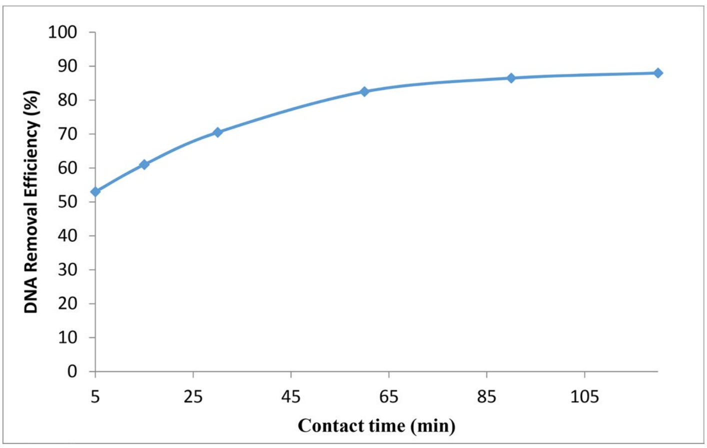


Figure 14. Effect of contact time.

DNA adsorption by adsorbent was not consistent, suggesting that electrostatic interaction between DNA and AIRSS was probably one of the involved mechanisms.

3.9. Contact time

To determine the effect of contact time between DNA ions and AIRSS, a batch experiment was carried out at various time intervals. The removal of DNA increases with increasing contact time, as seen in Figure 14. Since the adsorption of DNA ions in aqueous solution on the surface of AIRSS is equivalent to the desorption rate, there is no more rise in DNA adsorption after 120 minutes, suggesting that adsorption equilibrium was reached at 120 minutes, at which the DNA removal rate was constant.[35]

At different contact times, DNA reduction using AIRSS as an adsorbent is shown in Figure 14. DNA removal increased rapidly as contact time increased, but peaked at 60 minutes before stabilizing at a more or less constant value, suggesting equilibrium. DNA absorbs quickly, taking less than 60 mins. This may be because DNA was quickly adsorbed on the surface of AIRSS in an instantaneous sorption reaction. Furthermore, the rapid rise in DNA reduction (from 20 to 40 mins) shows that DNA adsorption is more likely triggered by absorption through the adsorbent's pores. After 60 minutes, the rate of DNA removal slowed significantly, suggesting equilibrium and the absence of sorption sites. Since no further important DNA reduction was observed after 60 mins. Here, the equilibrium cycle (60 mins) was chosen and used in all subsequent experiments. DNA ions rapidly bind to the readily available active sites on the outer surface of AIRSS, removing higher concentrations of DNA within the first 30 mins. Due to the rapid diffusion of DNA ions through the porous AIRSS's, DNA adsorption takes longer after 30 minutes. The adsorption process for DNA ion was observed even in a short period of time due to rapid initial stage adsorption, which was followed by a second stage with a comparatively slow adsorption process before equilibrium was reached. Equilibrium is reached when the percentage of DNA extracted remains constant despite the longer contact time as per the literature sources.[18] The reaction time for the supporting experiments was held at 120 mins because no substantial change in adsorption was observed after 120 mins in this analysis.

3.10. Adsorbent dose

The effect of changing the adsorbent dose on DNA absorption was studied at an optimum pH of 4.0, a contact time of 30 minutes, and an initial DNA ion concentration of 10 mg L^{-1}, as shown in Figure 15. The amount of adsorbent used has a significant impact on DNA absorption. The increased accessibility of a higher quantity of DNA ions for each unit mass of adsorbent causes the increase in adsorption ability with increased adsorbent volume. Owing to the saturation of the adsorbent surface sites, an increase in the adsorbent amount does not result in a substantial increase in DNA absorption.[36] With an adsorbent dosage of 0.4 g L^{-1} of DNA solution, the maximum percentage DNA removal was calculated to be 98.7%, which is a very good adsorption performance. As a result, the optimal dose of the adsorbent for further testing was determined to be 0.4 g L^{-1}.

The DNA adsorption increased with increasing the adsorbent dose from 1.0 g/L (45.6%) to 2.0 g/L (88.4%) for the adsorbent (Figure 15). For the

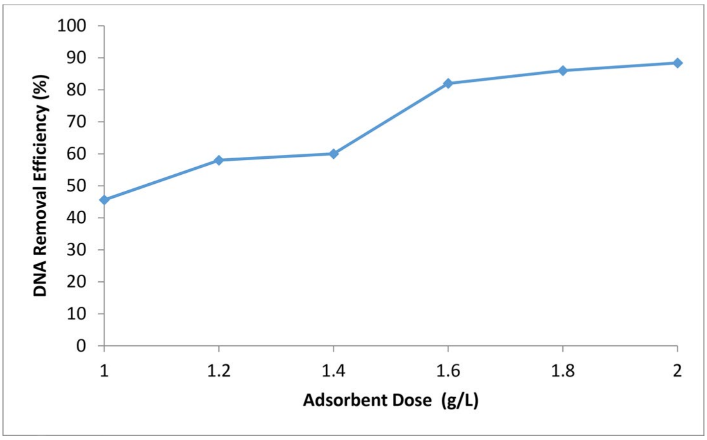


Figure 15. Effect of adsorbent dose.

sorption reaction, more sorbent surface was accessible at higher doses of the adsorbent, resulting in greater removal. It can also be noted that DNA removal increases initially as the dose increases, but there was no substantial increase in removal within a certain dose range. As a result shows that for the tea waste, the maximum removal efficiency of DNA was 88.4%.

Increases in DNA concentration above 3 mg/L, on the other hand, resulted in a 4% reduction in DNA elimination, most likely due to saturation of adsorption sites on the soil. As a result, the current experiment shows that if the concentration of AIRSS is less than 3 mg/L, it can eliminate greater than 88% DNA ions from water. The saturation activity of the adsorbent at these low DNA levels is likely attributable to the high ratio of surface active sites to overall DNA; hence, the DNA ions could interfere with the adsorbent surface to sufficiently occupy the active sites on the AIRSS surface, a significant factor favoring DNA removal. This finding showed that as the adsorbent dose was increased, the percent elimination increased due to an increase in the amount of possible active sites on the adsorbent surface before equilibrium was reached. When the removal efficiency does not improve significantly despite increasing the adsorbent dosage, equilibrium has been reached. Both types of sites are fully exposed for DNA adsorption at low dose for such adsorbents, and the surface becomes saturated faster, resulting in a higher qe value. The availability of higher energy sites decreases as the adsorbent dose increases, with a significant fraction of lower energy sites occupied, resulting in a lower qe.

3.11. Thermodynamic study

The function of temperature in the adsorption process was described using thermodynamics. The following equation was used to calculate thermodynamic parameters such as change in entropy (S), change in enthalpy (S), and change in Gibb's free energy (G) for the adsorption of DNA ions on adsorbent.

$$\mathrm{Kc} = \mathrm{q_e} / \mathrm{C_e} \tag{14}$$

$$\mathbf{ln\ Kc} = \boldsymbol{\Delta}\mathbf{S} / \mathbf{R} - \boldsymbol{\Delta}\mathbf{H} / \mathbf{RT} \tag{15}$$

$$\Delta\mathrm{G} = \Delta\mathrm{H} - \mathrm{T}\Delta\mathrm{S} \tag{16}$$

where K is the thermodynamic distribution coefficient constant, T and R denote temperature (K) and universal gas constant, respectively, in the reaction. C_e and q_e are DNA ions in the aqueous phase (mg/L) and adsorbed quantity of DNA (mg/g) on AIRSS at equilibrium.

The Gibbs free energy shift values for DNA ion adsorption on the adsorbent showed that the adsorption mechanism is spontaneous in nature, and that the spontaneity of the adsorption reaction increases with temperature. To assess the thermodynamic feasibility and spontaneity of the adsorption process, the thermodynamic parameters were determined (Table 7) from plots of log (qe/Ce) versus 1/T. At 333, 343, 353, 373 K, the adsorption process has a H° of 34.90 kJ mol^{-1} and a S° of −116.29 KJ mol^{-1}, resulting in G° values of 9.534, 9.720, 10.202, and −10.277 kJ mol^{-1}, respectively. The negative H° and S° values mean that an exothermic mechanism is dominant, and the total entropy of the system decreases. The adsorption mechanism is random and exergonic, as shown by the negative G°. As the temperature is increased from 333 to 373 K, the G° values become more negative, going from 9.02 to 10.277 kJ mol1, meaning that the reaction favorability increases with temperature, which is consistent with the existence of exothermic reactions that are entropically favorable.[18] The thermodynamic findings are consistent with kinetics studies that show an improvement in adsorption density and percent DNA elimination.

Thermodynamic parameters obtained from equilibrium isotherm results, such as the Gibbs free energy of adsorption (Gads), may also provide insight into the form and mechanism (physisorption or chemisorption) of the AIRSS-DNA interaction. The equation was used to calculate the Gibbs free energy of adsorption (Gads):

$$\left(\Delta\mathrm{G_{ads}}\right) = -\mathrm{RT}\ \ln\left(\mathrm{K}\right) \tag{17}$$

Table 7. Calculated values of thermodynamic parameters.

ΔH (kJ mol^{-1})	ΔS (kJ mol^{-1})	ΔG (kJ mol^{-1})			
		333 K	343 K	353 K	373 K
−34.90	−116.29	−9.534	−9.720	−10.202	−10.277

Table 8. Thermodynamic data on DNA adsorption dynamics onto AIRSS.

DNA (mg L^{-1})	ΔH (kJ mol^{-1})	ΔS (kJ mol^{-1})	ΔG (kJ mol^{-1})		
			298 K	308 K	318 K
2	−60.35	−0.180	−8.120	−5.800	−5.432
3	−78.10	−0.265	−8.108	−4.110	−3.400
4	−70.60	−0.220	−9.200	−5.750	−5.540
5	−45.44	−0.136	−7.600	−5.464	−5.210
6	−50.22	−0.150	−8.106	−6.220	−6.216
7	−56.45	−0.165	−8.700	−5.650	−5.320
8	−40.50	−0.104	−8.200	−6.300	−6.160

where K is the equilibrium constant, related to the Langmuir isotherm constant (b), R is the molar gas constant (8.134 J/mole), and T is temperature in Kelvin.

$$K = 55.5 \times b \tag{18}$$

The DNA-AIRSS interaction is random, according to a negative G_{ads} value (= −28.44 kJ/mole) obtained in this analysis. The G_{ads} value for physical adsorption is typically in the region of −20 kJ/mole, whereas that for chemisorption is typically in the range of −80 to −400 kJ/mole. As a result, the other order of magnitude of the G_{ads} (= −28.44 kJ/mole) indicates that the DNA adsorption on AIRSS was a physico-chemical process.

In this case, two types of forces can cause sorption: enthalpy-related and entropy-related forces. The enthalpy-based forces are greater in hydrophilic bonding due to the additional contribution of electrostatic interactions. The isosteric enthalpy of DNA sorption in the current study ranged from 43.65 to 95.18 kJ mol^{-1}, suggesting an exothermic process or exothermic binding reaction. A chemical adsorbs to a solid adsorbent when the free energy of the sorptive exchange is negative. The negative G values, which decrease from 298 to 318 K, indicate that DNA sorption is spontaneous. Physical forces govern bonding because the current DNA sorption's free energy change was calculated in the range of 0–20 KJ mol^{-1}. The feasibility of DNA sorption decreases from 298 K to 318 K, as shown in Table 8.

Lower DNA sorption at higher temperatures could be attributed to the temperature dependence of sorption coefficients and solubility. As a result, the temperature influence on sorption isotherms is equal to the total of the contributions from sorption and solubility. The decreasing sorption at higher temperatures may be due to weak interactive forces between AIRSS and DNA at 500 °C, suggesting physisorption. The negative entropy change during DNA sorption at all temperatures means that the system is orderly and enthalpy driven. The entropy transformation reflects the decreased mobility of adsorbed DNA compared to that in solution.

3.12. Regeneration study

Regeneration experiments are concerned with the adsorbent reusability, which reduces operational expenses and protects the environment. Conventional acids and bases are used in the regeneration process. The regeneration ability of adsorbent was conducted with HCl or NaOH. During the treatment, the HCl can cause metal component leaching and adsorbent damage, whereas the most powerful solution for adsorbent regeneration was found to be NaOH, which favors the substitution of DNA. The rubber seed shell waste adsorbent (AIRSS) was performed using batch experiments. In this process, a lower concentration of NaOH was chosen and regenerates AIRSS because higher concentrations of NaOH can modify the structure of adsorbent.

Adsorbent with a concentration of 2 g/L was used in this experiment, and it was stirred with sodium hydroxide solution for 120 minutes. The adsorbent was then filtered and dried for 6 hours at 95 °C in an oven. The regenerated adsorbent was used in the removal of DNA from water at all levels. In the tenth regeneration cycle, chemisorption of previously adsorbed DNA that was not completely removed from the adsorbent during the desorption process, as shown in Figure 16, reduced adsorption efficiency to 52%. There is a chemical reaction between the AIRSS and DNA during the chemisorption phase, making the regeneration process difficult and that the adsorbent's reusability.

When the regenerated AIRSS was compared to other adsorbents in literature sources, it was discovered that AIRSS has a higher reusability than other adsorbents. Because of its microporous composition and higher surface area, AIRSS can be reused, lowering the overall cost of the adsorption process in practical applications.

3.13. Adsorption mechanism

The removal of DNA by activated carbon is primarily governed by a physical adsorption process, with surface area playing a key role. Furthermore, it was discovered that the intrinsic functional groups on the carbonaceous material surfaces, such as phenolic hydroxyl groups and carboxyl groups, had no effect on the removal of DNA ions from aqueous

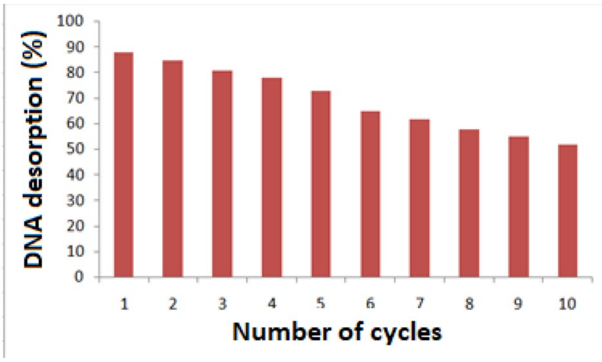


Figure 16. Regeneration cycle.

solution. Carbonaceous materials showed low DNA adsorption efficacy which was attributed to their nonmetallic existence as compared to carbonaceous materials modified with metallic compounds like alumina, which have a high preference for aqueous DNA ions.

The adsorption mechanism used by AIRSS to remove DNA is based on the Langmuir isotherm model of pseudo-second-order fitting. Electrostatic or/and ion exchange between hydroxyl groups and DNA ions is the mechanism for DNA adsorption. The zeta potential value of $Al(OH)_3$ was found to be 45.2 mV in aqueous solution (pH 4.0), demonstrating that the surface of $Al(OH)_3$ in neutral solution was positively charged. Since DNA ions are negatively charged, electrostatic attraction was used to bind the anions to the adsorbent's surface.

3.13.1. UV-Visible spectrum

Using electronic absorption spectroscopy, the interactions between adsorbent (AIRSS) and DNA were established. DNA binds with metallic ions through electrostatic finding mode due to cationic species and phosphate groups of DNA. The electronic absorption spectra of adsorbent (AIRSS) in the absence and presence of DNA are shown in Figure 17. In the absence of DNA, the adsorbent (AIRSS) showed a band at 262 nm which corresponds to $\pi \rightarrow \pi^*$ transition. This absorption band was experienced both hyperchromism and bathochromism due to the addition of DNA, indicates that the complexes bind to CT-DNA through electrostatic binding mode. There is a change in wavelength and absorbance at 520 nm which indicates the binding of aluminum ion and the phosphate group of DNA (Figure 17). The intrinsic binding constant (K_b) of adsorbent with DNA was determined based on the absorbance and molar extinction coefficient. In order to compare the binding strengths of the adsorbent with CT-DNA, the K_b value is 3.60×10^5 M^{-1}, respectively. It is suggested that the binding

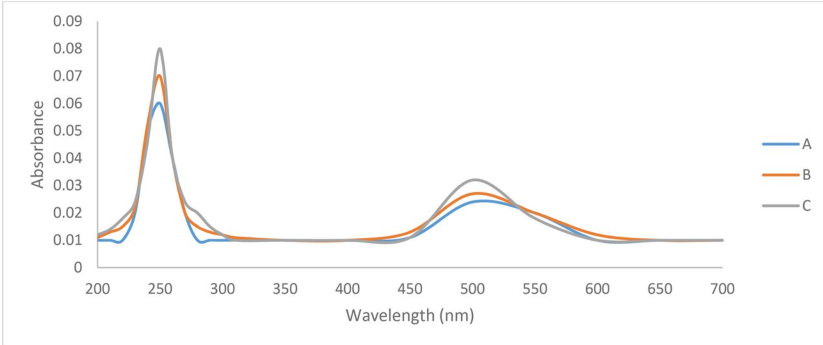


Figure 17. UV-Visible spectrum of adsorbent (AIRSS) alone (A) and with different concentrations of DNA (B, C).

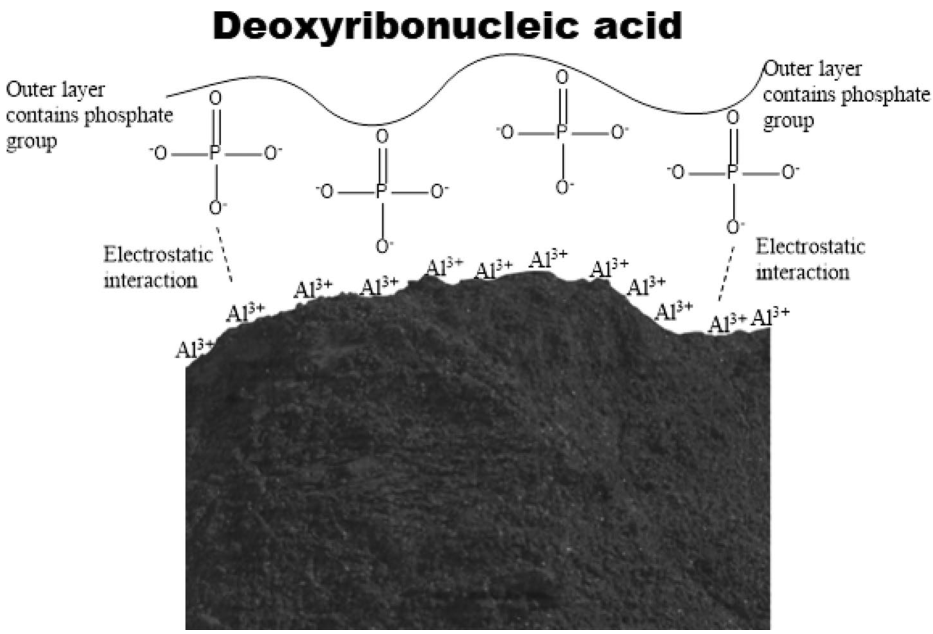


Figure 18. Pictorial representation indicates that electrostatic interaction occur between adsorbent surface aluminum ion and phosphate group of DNA.

strength of copper(II) complex to DNA is stronger than that of ethidium bromide (EB) to DNA ($K_b = 3.3 \times 10^5$ L·mol^{-1}) due to the peripheral aluminum ion (Figures 18 and 19).

In the spectrum, hyperchromism (increased absorption) or hypochromism (decreased absorption) can be seen (decreased in absorption). The adsorbed DNA molecule is more easily separated from the adsorbent and extracting efficiency is also higher as compared to intercalation binding mode. The free energy change was also calculated for the above-said binding of DNA and found to be 52 Kcal. M^{-1}.

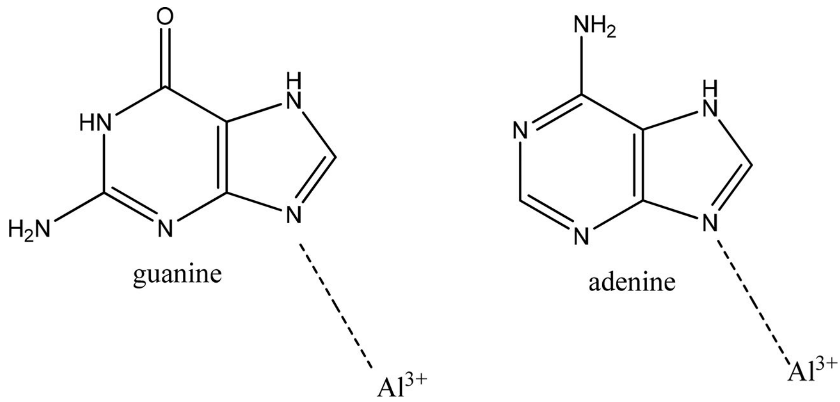


Figure 19. Pictorial representation indicates specific interaction of aluminum ion with guanine and adenine residue in the DNA through intraparticle diffusion.

We conclude that aluminum ions binds with CT DNA through electrostatic binding nature under low pH conditions and precipitate occurred. Aluminum is an essential element and may be due binding to DNA. Aluminum is covalently bound to the N7 of guanine and adenine (Figure 20). These structural features reveals that the aluminum interacts with biological systems and provide a research benefits in human health (extraction of DNA).

3.14. Cost–benefit analysis

The adsorbent's cost–benefit analysis is a critical factor in determining its economic viability. AR-grade chemicals and agrowaste materials were used to create the formed nanoporous AIRSS. The cost–benefit analysis revealed that mesoporous AIRSS has a lower DNA removal cost than other conventional adsorbents. After ten adsorption cycles, the nanoporous AIRSS had a higher removal of DNA. As a result, the low-cost, smart, and long-lasting material (nanoporous AIRSS) demonstrated strong DNA removal efficacy in water.

Figure 20. DNA removal using AIRSS stored at different times.

3.15. Effect of storage time on the DNA removal capability of AIRSS

Batch adsorption experiments using freshly generated RSS and mesoporous AIRSS stored for 8 months under normal room conditions (temp. = 25 °C) were used to determine the effect of storage time on the efficiency of removal of DNA. After 9 months of storage, the DNA removal efficiency of mesoporous AIRSS decreased (Figure 20). The factors that led to the decreased DNA removal efficiency were not investigated further. However, preliminary findings indicate that storing mesoporous AIRSS for long periods of time after processing before use is not recommended. The output of AIRSS remained nearly unchanged after 9 months of storage, indicating that the aluminum (hydr)oxides were sufficiently bound to the RSS surfaces and underwent no major transformations under storage conditions (room conditions) that could affect its quality.

3.16. Leaching test for utilized AIRSS for safe discharge into landfill

The selected adsorbent, AIRSS, was used to remove DNA ions from water supplies in the current research. After the results, the adsorbent was safely disposed of in a landfill with no harmful effects on the environment. In the micro and mesoporous positions, the adsorbent was bound with DNA ions. Aluminum and DNA ions are present in the adsorbent as it dissolves in water. Aluminum (Al) is the most abundant metal and the third most abundant element in the Earth's crust, in general. Solubilization of Al from rock ore increases as soil pH decreases, resulting in an accumulation of soluble ionic Al (Al^{3+}) in strongly acidic soils. Organic acid anions, phenolic compounds, and polysaccharides in root mucilage all combine to form nontoxic Al complexes in plants. The beneficial effects of Al are thought to be attributable to increased uptake of vital nutrients (e.g., phosphate) or relief from the effects of other harmful factors (e.g., Fe^{2+} and/or H^+ toxicity). Al^{3+} is needed for root growth and development in tea plants, according to research. Similarly, DNA prevents dental caries, aids in the formation of dental enamels, and prevents bone mineralization deficiencies when ingested of sufficient amounts in water. A systematic toxicity characteristic leaching procedure was employed and find out the DNA concentration in the utilized adsorbent sample (AIRSS) and presented in Table 9.

4. Conclusion

Alumina surface modification of agro waste material, a locally accessible material in developing countries, was found to be useful for contaminants from water sources. In this study, DNA removal from water was

Table 9. Toxicity characteristic leaching procedure-extraction from spent AIRSS.

Spent AIRSS	Average DNA concentration (mg/L) in triplicate samples
Untreated (without DNA stabilization)	10.52 ± 0.20
Treated with Al (DNA stabilized)	1.18 ± 0.10
WHO regulatory level at Room temperature	120

determined using aluminum impregnated rubber seed shell adsorbent. The surface morphological features, porosity and adsorption performance for DNA was thoroughly explored. The proximate analysis indicates the composition of ash, volatile, moisture and fixed carbon content and ultimate study gave elemental composition (C, S, N and H) of the adsorbent AIRSS. Surface texture and morphological features of adsorbent was confirmed with the aid of SEM. Due to its porous structure and predominately micro porous character, the rubber seed shell waste adsorbent has a high surface area and adsorption ability. The crystal phase, structure and average grain size of the adsorbent is determined by X-ray diffraction technique. The result gave the average crystalline size was 53 and 48 nm for before and after adsorption with adsorbent. The adsorption percentage increased with increasing contact time and adsorbent dose, according to the reaction process conditions. Pseudo-second order chemical reaction kinetics provided the strongest association of experimental results for AIRSS. The Langmuir model is effective in eliminating DNA from AIRSS, as shown by the correlation coefficient 3.909 mg/g is the maximum adsorption capability. Adsorbent materials derived from agro waste materials have complex organic molecules based on soil composition. In adsorbent products, the above determines the identification of chemical entities. Adsorbent effectiveness is determined by factors such as adsorbent concentration, temperature, pH, and contact time. DNA removal improved as adsorbent dosage, pH, temperature, and contact time were increased. F^- adsorption decreases with rising temperature and increases with increasing absorbent dose, according to the findings. Al^{3+} is classified as a hard acid due to its properties, whereas F^- is classified as a hard base. According to the HSAB principle, Al^{3+} has a high affinity for DNA, which is why it is used to modify the surfaces of native materials as an alternative material for DNA adsorption. At 2 g/L, pH 4, and a reaction time of 120 minutes, the maximum percentage (92.5%) of DNA was removed. After 9 months of storage under standard room conditions, the DNA removal efficiency of AIRSS was found to be reduced. Adsorption of DNA into AIRSS was studied by evaluating the relationships between cation identity and concentration, pore diameter, and surface area with respect to DNA adsorption. Before and after adsorption, XRD and N_2 physisorption experiments revealed that the DNA was localized inside the pores rather than on the exterior surface. The type of metal cation employed to mediate the charge between the DNA molecule and the silica surface was discovered to affect DNA adsorption, with DNA having a higher affinity for Al^{3+}. The amount of DNA that could be put into AIRSS was also shown to be affected by the pore diameter. DNA adsorption was shown to be more favorable in materials with pores greater than 50, since the molecules could likely penetrate the pores without significant intermolecular interactions. The mesoporous AIRSS was utilized as an adsorbent and found to be promising for removing DNA from contaminated groundwater while also contributing to long-term sustainability.

Disclosure statement

No potential conflict of interest was reported by the authors.

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